Advanced Machine Learning, ETH Zurich

Diffusion models

The forward and backward process, training, and conditioning

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Subject

Which are real? Which are fake?

Entry

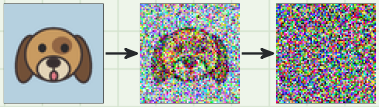
1



W. Peebles and S. Xie, arXiv:2212.09748, Fig. 30, cropped. CC BY 4.0.

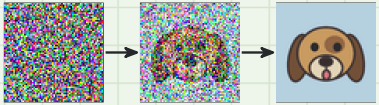
A diffusion model generates new data, such as images, by learning to undo a process that gradually adds noise.

Training



Take real images, add noise in many small steps, and learn to predict the noise that was added.

Generation

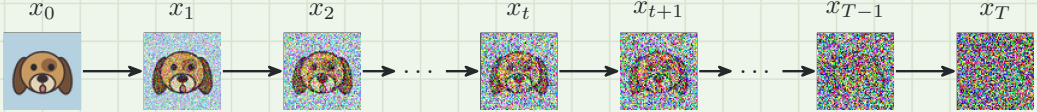


Start from pure noise and remove the predicted noise, step by step, until a clean image appears.

Each forward step fades the image and adds noise; generation runs the chain backwards.

fade & noise

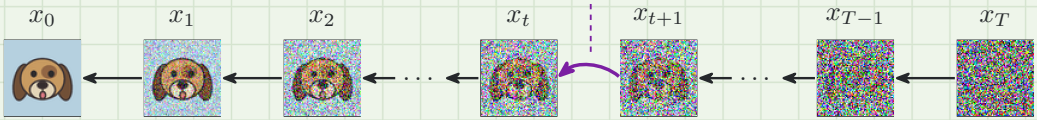
$\varepsilon_0 + \varepsilon_1 + \varepsilon_2 + \varepsilon_{t-1} + \varepsilon_t = \hat{\varepsilon}_t$



If you knew $\hat{\varepsilon}_t$ then you could approximate well x_t from x_{t+1}.

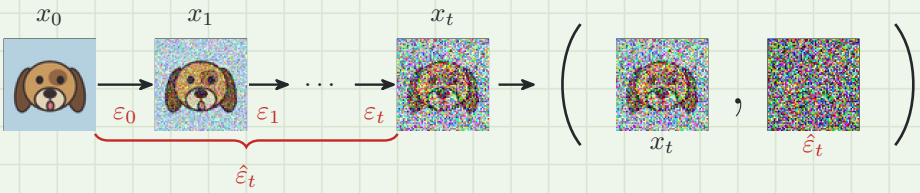
$=$

remove noise



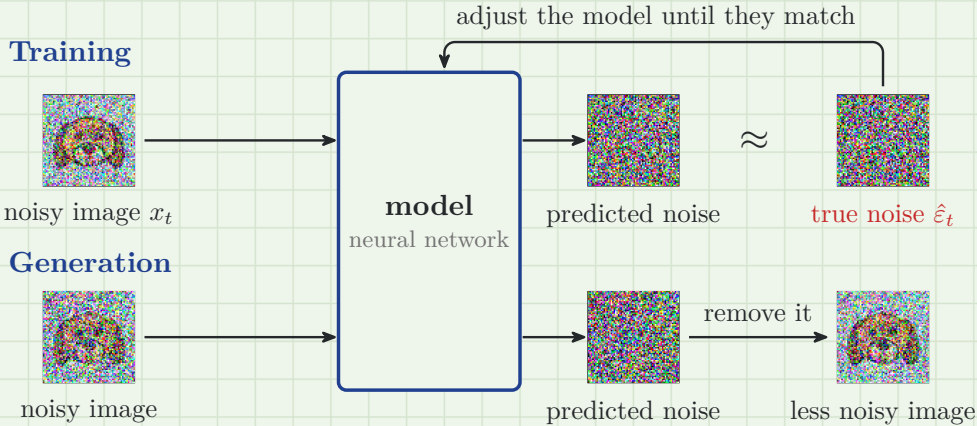
The model is trained to predict the noise $\hat{\varepsilon}_t$ from the noisy image x_t.

1. Take an image of the dataset.
2. Pick $t \leq T$ at random.
3. Forward t steps.



Repeat until you collect $\{(x_t^{(1)}, \hat{\varepsilon}_t^{(1)}), \ldots, (x_t^{(n)}, \hat{\varepsilon}_t^{(n)})\}$

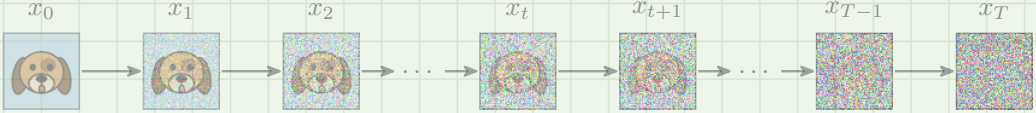
One neural network, two uses: trained to predict the noise, then used to remove it.



The prompt "cat" is given to every backward step, so the image it produces is a cat.

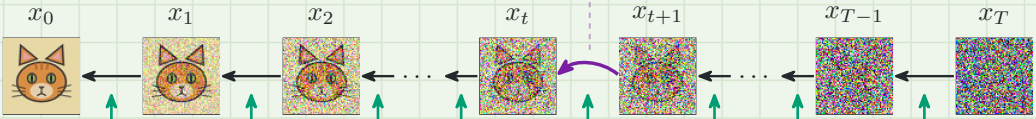
fade & noise

$\varepsilon_0 + \varepsilon_1 + \varepsilon_2 + \varepsilon_{t-1} + \varepsilon_t = \hat{\varepsilon}_t$



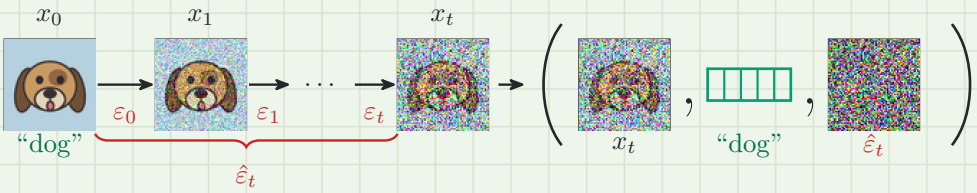
If you knew $\hat{\varepsilon}_t$ then you could approximate well x_t from x_{t+1}.

remove noise



Each training image comes with its caption, which is fed to the model as well.

1. Take an image of the dataset, with its caption.
2. Pick $t \leq T$ at random.
3. Forward t steps.



Repeat for many captioned images.